LAB 5

UML Sequence & Activity Diagram Modeling

Digital Document Verification System

Aim

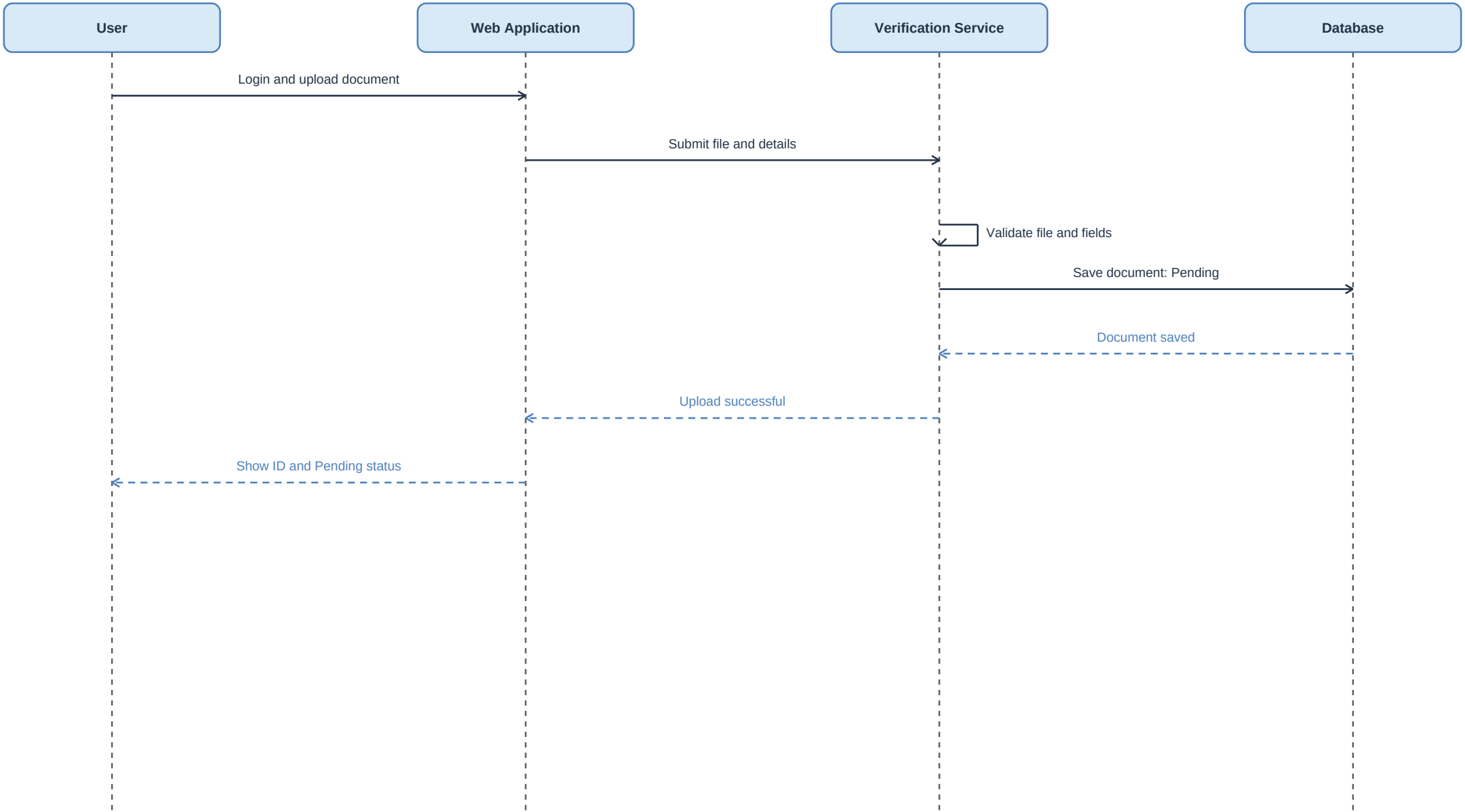
To model major scenarios of a Digital Document Verification System using UML Sequence Diagrams to show message flow and an Activity Diagram to show the workflow.

Main actors and components

User	Uploads documents and checks verification status.
Web Application	Provides the interface for users and administrators.
Verification Service	Validates documents and updates verification status.
Database	Stores document metadata and verification results.
Admin / Verifier	Reviews documents requiring verification.

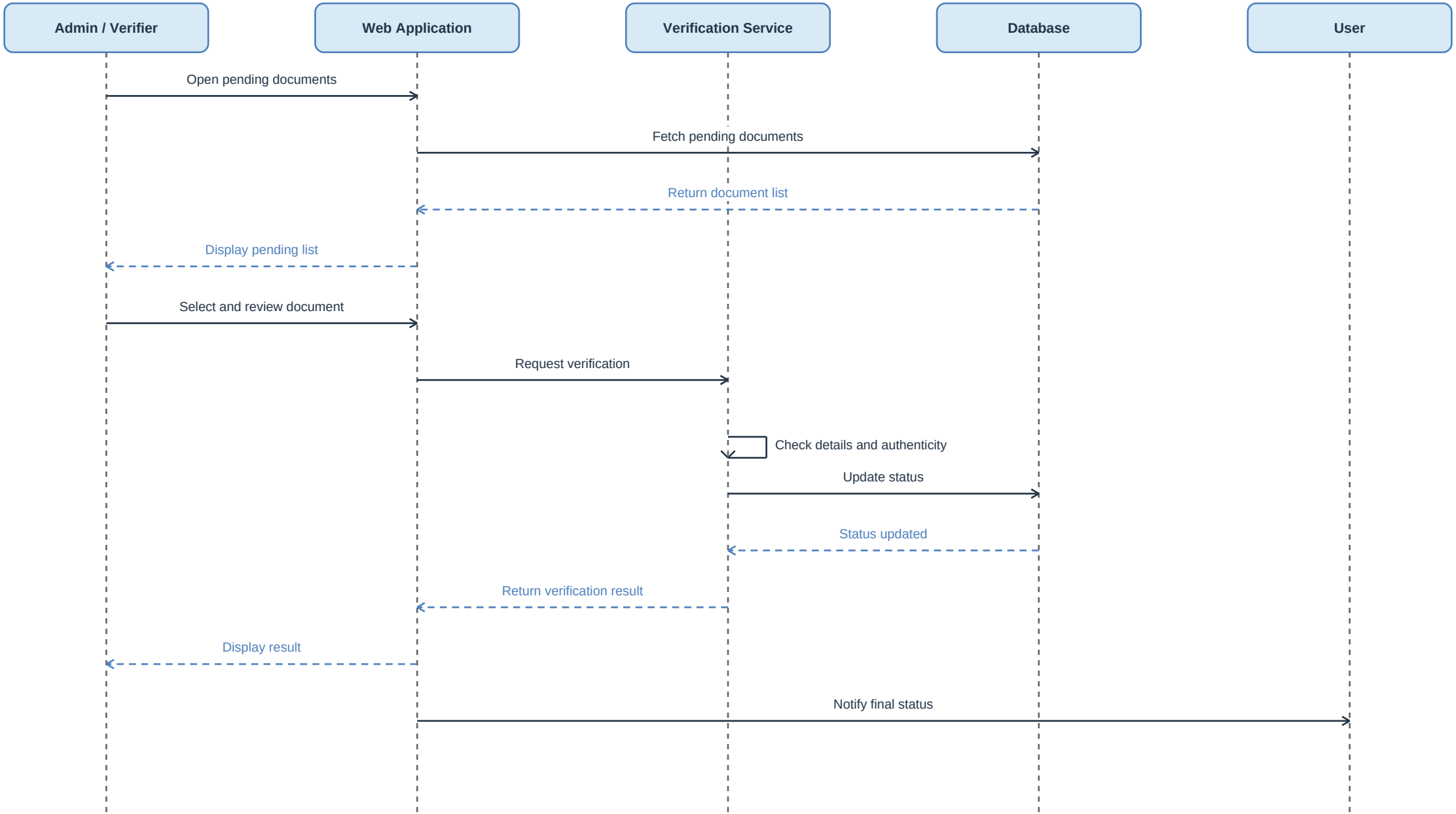
Sequence Diagram 1 — Upload Document

Successful upload stores a valid document as Pending; invalid files return an error.



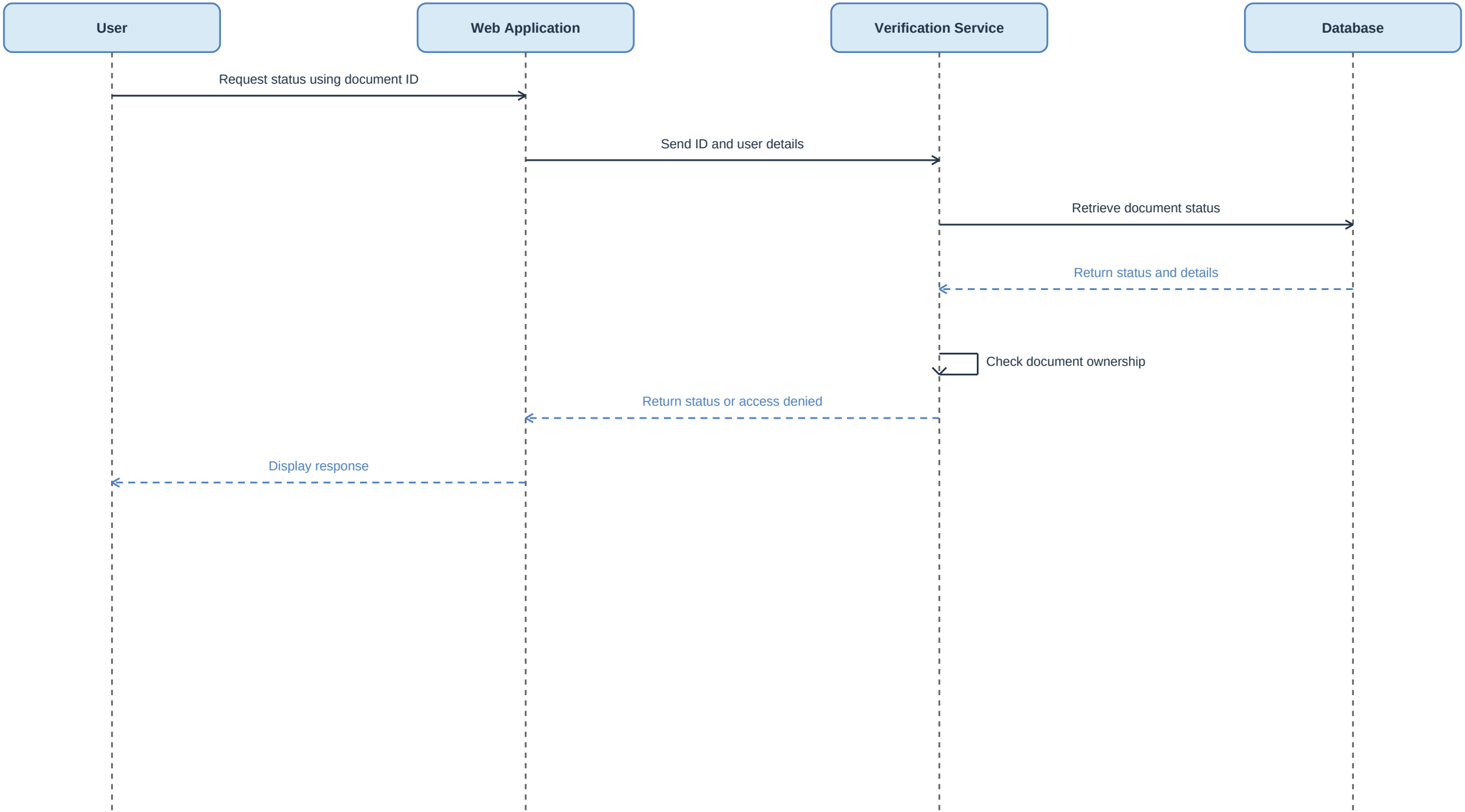
Sequence Diagram 2 — Verify Document

An admin reviews a pending document; the system records Verified or Rejected.



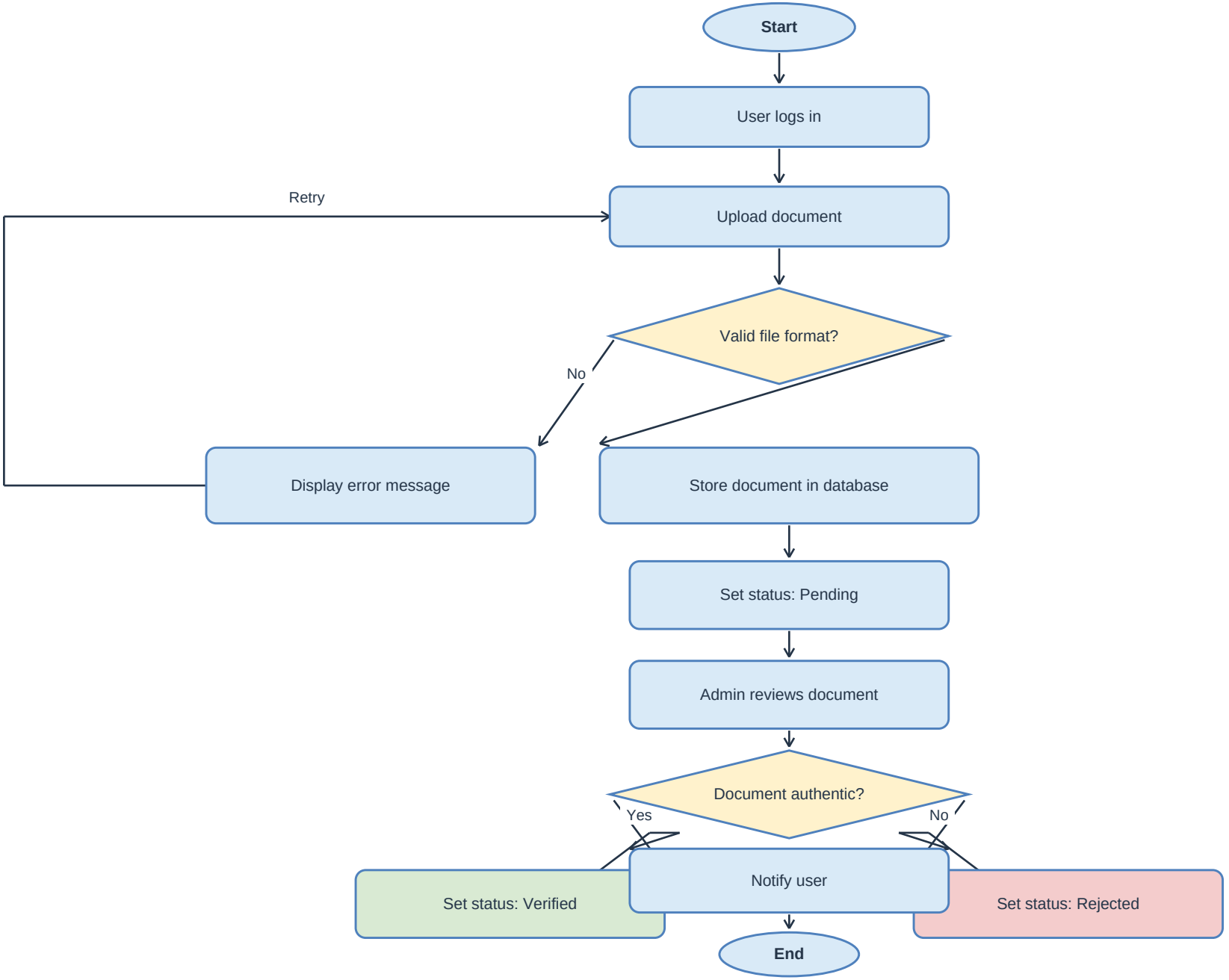
Sequence Diagram 3 — Check Status

The service checks document ownership before returning the current status.



Activity Diagram — Verification Workflow

Draw.io-style flowchart using start/end nodes, activity boxes, decision diamonds, and labeled connectors.



Result and Conclusion

Result

The UML Sequence Diagrams and Activity Diagram for the Digital Document Verification System were modeled. The sequence diagrams show interactions among users, the web application, verification service, database, and admin. The activity diagram shows the workflow from login and upload to final status notification.

Conclusion

UML modeling helps visualize system behavior before implementation. Sequence Diagrams describe communication between components, while Activity Diagrams represent activities, decisions, and alternative paths in the verification process.